HandRT: Simultaneous Hand Shape and Appearance Reconstruction with Pose Tracking from Monocular RGB-D Video

Pratik Kalshetti, Parag Chaudhuri

***Abstract*—We propose a method to reconstruct a personalized hand avatar, representing the user's hand shape and appearance, from a monocular RGB-D video of a hand performing unknown hand poses under unknown illumination. Our method, HandRT, jointly optimizes hand pose, shape, appearance, and lighting parameters using a physically-based shading model in a differentiable rendering framework incorporating Monte Carlo path tracing. HandRT extends our previous work, Intrinsic Hand Avatar, by relaxing the assumption of a known coarse hand pose and utilizing depth data in the optimization. Specifically, we introduce an articulated registration energy based on iterative closest point over the depth point cloud that enables reconstruction from unknown hand poses via tracking across frames. Thus, we can reconstruct the avatar from arbitrary poses with high accuracy without relying on an off-the-shelf 2D joint detector at each frame. Further, HandRT is capable of precisely tracking the reconstructed avatar from either RGB or RGB-D input. Our evaluation demonstrates that our method outperforms existing hand avatar reconstruction methods on all commonly used metrics while producing significantly accurate mesh compared to state-of-the-art hand mesh recovery methods by a large margin on public and our captured datasets. To facilitate future research, we release our code at https://github.com/pmkalshetti/handrt.**

***Index Terms*—Hand shape and pose estimation, appearance reconstruction, hand tracking.**

I. Introduction

A user-specific digital hand avatar is essential to realize a realistic and immersive experience in virtual reality (VR) and augmented reality (AR) applications. In our context, an avatar is represented by geometry (e.g., triangle mesh) and appearance (e.g., spatially-varying material). The geometry of a hand is further decomposed into shape and pose: the shape captures identity-specific parameters in a neutral/rest configuration of the hand mesh (e.g. a stretched hand), while the pose captures the articulation out of the rest configuration. Further, the realistic appearance of the user's hand texture plays a vital role in providing a sense of embodiment, whereas accurate modeling of the user's hand geometry and tracking is essential for precise interaction with virtual objects. We focus on the problem of automatically reconstructing such a personalized hand avatar from a monocular RGB-D video while tracking its articulated pose to resemble the user's hand shape, appearance and pose.

Recently, there has been a lot of research in creating digital avatars of humans [12, 16, 17, 55, 67] and faces [13, 79]. However, reconstructing and tracking hands from image observations has unique challenges, like substantial pose variation among similar-looking fingers and self-occlusion compared to full-body and face. Further, hands play a much more critical role in an interactive experience, so developing methods tailored for reconstructing and tracking hand avatars is essential.

The importance of a user-specific (or calibrated) hand shape model for accurate pose tracking is well-established in the literature [58, 59]. Traditional hand tracking methods use iterative closest point (ICP) style algorithms to register a hand model to a sequence of depth images [58, 59]. However, these methods rely on the inaccurate silhouette obtained from noisy depth data [53] and do not utilize RGB information. With the advent of data-driven blend-shape models for hands [20, 49], most hand shape and pose estimation methods either directly regress or optimize these parameters from RGB [6] or depth images [19]. However, these methods are limited by the data-driven hand model, which cannot adapt to unseen hand shapes with significant deviations from the training set and thus adversely affects tracking. To tackle this, recent methods [27, 31, 32] directly regress mesh vertices from the input image. However, these learning-based methods fail to generalize to unseen data.

More recently, modeling hand appearance together with shape has gained attention [7, 10, 22, 30, 47]. Traditional approaches to hand appearance modeling learn a PCA basis of textures [30, 47] from a large scan of various hands. However, these approaches cannot generalize to unseen hand appearances. Recent advances in neural rendering have enabled learning digital hand avatars from videos [7, 10, 22]. However, these methods rely on an existing data-driven hand mesh recovery method [31] for pose initialization, eventually leading to blurry textures due to registration mismatch. Apart from the assumption of a known pose, these approaches constrain the scene (e.g., single light source, multi-view setup) and require considerable training times.

We propose a novel method, *HandRT*, to reconstruct a personalized hand avatar from a monocular RGB-D video of a hand performing unknown hand poses under unknown illumination (see Figure 1). Our avatar representation comprises a parametric triangle mesh and spatially-varying physically-based materials [34]. We model the illumination using a high dynamic range environment light probe to disentangle the intrinsic appearance of the user's hand and illumination. Our method jointly optimizes a parametric mesh, material, and lighting using differentiable rendering based on Monte

Pratik Kalshetti and Parag Chaudhuri are with the Department of Computer Science and Engineering, Indian Institute of Technology Bombay.
E-mail: pmkalshetti@iitb.ac.in

Fig. 1. We propose a novel method, *HandRT*, to reconstruct the personalized hand shape and appearance of a user's hand captured under real-world environment lighting while simultaneously tracking the articulated pose from a monocular RGB-D video.

Carlo path tracing. Our reconstructed hand avatar captures realistic shading (c.f. HandAvatar [7]) under more general real-world environments (c.f. HARP [22]). Unlike existing methods [7, 21, 22] that assume a known coarse hand pose, we relax this assumption by utilizing depth data in the reconstruction optimization. To this end, we introduce an articulated registration energy based on iterative closest point (ICP) over the depth point cloud that enables reconstruction from arbitrary unknown hand poses via tracking across frames. Further, our tracking framework directly optimizes using only the input RGB-D video without the need for any annotations, and thus, by design, generalizes to new data. We compare our method with state-of-the-art avatar reconstruction methods on a publicly available dataset and observe that it outperforms them on all commonly used metrics. We also compare our method with state-of-the-art hand mesh recovery methods and observe that our method reconstructs significantly accurate mesh by a large margin on public and our captured datasets.

Relation to Intrinsic Hand Avatar [21]: This paper extends our previous conference paper, Intrinsic Hand Avatar [21], which reconstructs a hand avatar from monocular RGB video, in the following ways.

1) This paper relaxes the primary assumption in Intrinsic Hand Avatar of a known coarse hand pose and utilizes depth data in the optimization. Specifically, we introduce an articulated registration energy based on iterative closest point over the depth point cloud (see Section III-C1). This enables us to reconstruct the avatar from arbitrary unknown hand poses via tracking, unlike Intrinsic Hand Avatar which estimates coarse hand pose from 2D joint detection. Further, the depth data also helps in optimizing the shape and appearance in the joint optimization leading to more accurate avatar reconstruction.
2) Further, this paper provides extensive experimental evaluation (see Section IV-C) by comparing with state-of-the-art RGB- and depth-based hand tracking approaches. We also quantitatively compare HandRT and Intrinsic Hand Avatar in Table V.

II. Related Work

Recovering human hand motion and appearance is a widely studied problem in computer graphics and vision. In this section, we briefly review the literature on reconstructing hand shape and appearance, and pose estimation from RGB or depth or RGB-D images.

A. Hand Tracking

RGB-based methods: Gorce *et al*. [11] introduced the idea of analysis-by-synthesis to estimate the full hand surface represented by a triangle mesh. With the advent of the parameterized 3D hand mesh model, MANO [49], hand shape and pose estimation from monocular images has rapidly progressed. Boukhayma et al. [6] regress the hand model parameters from RGB image by utilizing 2D/3D joints as weak supervision. Baek et al. [2] rely on a neural renderer and use a segmentation mask for additional supervision. To further improve the accuracy, Kulon et al. [27] use a GraphCNN to optimize directly on mesh vertices obtained using a parametric model fitting approach. Zimmermann et al. [80] introduce a semi-automatic model fitting approach to generate more reliable mesh annotations on RGB images. Current state-of-the-art mesh reconstruction methods [31, 32] utilize this data in advanced network models and learning strategies. Recently, ACR [71] introduced an attention collaboration-based regressor capable of handling two-hand interaction while achieving comparable accuracy on single-hand reconstruction. However, these learning-based methods fail to generalize on unseen data. On the contrary, our self-supervised method does not need any annotations and can directly optimize using only the input video and significantly outperform existing state-of-the-art hand mesh recovery methods.

1) Depth-based methods: One of the seminal works on depth-based hand tracking, proposed by Oikonomidis *et al*. [41], used a primitive-based geometrical model for optimizing the pose using particle swarm optimization (PSO). Makris and Argyros [33] further extended it by incorporating an online shape adaptation of the hand model. Tagliasacchi *et al*. [57] revived the idea of using ICP-like algorithms for hand tracking by fitting a cylinder-based hand model using Levenberg-Marquardt (LM) non-linear least-squares optimization. However, these methods still use a primitive model that lacks expressivity compared to our mesh-based model. We adapt these energy terms to be used with a surface mesh model in our proposed tracking framework.

Taylor *et al*. [59] introduced user-specific hand mesh, and Sharp *et al*. [52] used it in their robust hand tracker. Khamis *et al*. [24] learned a shape space of hand from depth images of multiple users, which enabled Tan *et al*. [58] to extend their golden-energy tracking method to various users. The notable work of Taylor *et al*. [60] used a subdivided hand mesh to track using the LM algorithm. Further, Shen *et al*. [53] demonstrated that one could get away without subdividing the mesh using ideas from Phong shading. Unfortunately, unlike ours, these implementations are not publicly available, restricting their usage for scientific research.

Most tracking algorithms rely on a robust per-frame discriminative technique to avoid drifting. The early work by Qian *et al*. [46] introduced the idea of robust fingertip detection that exploits finger geometry. With the availability of large scale hand pose datasets(e.g. NYU [62], MSRA [56], Bighand2.2M [72]) with 21 annotated 3D landmark points (16 joints and 5 fingertips), state-of-the-art hand pose estimation networks [15, 64, 68, 77] regress the 3D landmark points directly from depth images. However, all of these above methods focus on the user's hand shape and pose, and do not capture its appearance. Our geometric model leverages a parametric hand mesh model to regularize the optimization while allowing modification to the mesh to capture fine-level details specific to the user, and also captures user-specific appearance.

B. Shape and Appearance Reconstruction

Inspired by 3D morphable face model (3DMM) [4], HTML [47] proposes a linear appearance model for inferring UV texture, and NIMBLE [30] infers an albedo, specular and normal map. However, these methods are limited by their training set and thus cannot capture unseen hands whose appearance lies outside their dataset. Our method directly infers the intrinsic color of the user's hand from image observations and thus generalizes better than existing PCA-based textures.

The recent success of neural implicit representations has led to several methods reconstructing 3D geometry and appearance from image collections. NeRF [36] and follow-ups [42, 55, 70] use a volumetric representation to calculate the radiance accumulated along a ray by marching through a 5D light field. However, these methods fail to capture accurate geometry because of the ambiguity of volume rendering. Implicit surface reconstruction methods(VolSDF [69], UNISURF [40], NeuS [65]) leverage an occupancy network or signed distance function to model a more accurate geometry. Apart from the computationally expensive ray marching strategy for rendering, these methods cannot handle dynamic scenes. To handle deformable shapes, SNARF [8] proposed forward deformation fields from canonical space to deformed space and uses iterative root finding to find correspondences. SelfRecon [16], InstantAvatar [17] uses this idea to reconstruct human avatars, while IMAvatar [79] constructs face avatars from monocular video. LISA [10], the first method to learn an implicit shape and appearance of hands from multiview images, uses kinematic transformations of the bones to handle deformations. While these implicit surfaces can be converted to meshes for traditional graphics applications, this is suboptimal.

More recently, some methods [12, 78] combine explicitly mesh-based representation with deformation networks and appearance networks for compatibility with standard graphics pipelines. However, none of the above methods handle lighting, which leads to incorrect shadows and baked-in illumination in the albedo. HandAvatar [7] attempts to mitigate this problem by introducing a self-occlusion-aware shading field which comprises an albedo field and an illumination field. However, their network requires a large training dataset per user per environment, and thus inference suffers on out-of-distribution data. HARP [22] avoids any neural networks by using an analysis-by-synthesis approach to model hand shape and appearance. Specifically, it uses a mesh-based parametric model, a vertex displacement map, a normal map, and albedo together with a shadow-aware differentiable rendering to obtain a personalized hand avatar from monocular video. However, it can only handle a single light source and has limited texture resolution. Even more recently, MANUS [44] represented the hand using articulated 3D Gaussians [23]. However, their method relies on a multi-view input capture setup and uses a large number of Gaussian primitives (300K) making it computationally expensive. Our method also uses an analysis-by-synthesis approach to generate a personalized hand avatar from monocular video, but can handle environment lighting and produces a more detailed texture compared to HARP and MANUS, and generalizes to new shapes and appearances, unlike HandAvatar.

C. Material and Lighting Estimation

Reconstructing an object from images is the problem of inverse rendering, which decomposes the underlying intrinsic properties such as geometry, material and lighting. The forward rendering equation [18] (for non-emissive surfaces) computes the outgoing radiance L_o at surface point $\mathbf{x}$ along direction ω_o by integrating the reflected light over hemisphere Ω:

$$L_o(\mathbf{x}, \omega_o) = \int_{\Omega} L_{in}(\mathbf{x}, \omega_i) f(\mathbf{x}, \omega_i, \omega_o)(\omega_i \cdot \mathbf{n}) \mathrm{d}\omega_i \quad (1)$$

where $L_{in}(\mathbf{x}, \omega_i)$ is the incoming radiance along direction ω_i and f denotes the bidirectional reflectance distribution function (BRDF) which describes how much light arriving from direction ω_i at $\mathbf{x}$ is reflected towards ω_o.

Recent methods have successfully estimated lighting and BRDF from image collections. To obtain this intrinsic decomposition, NeRD [5] and PhySG [73] use spherical Gaussians, NeRFactor [75] uses a low-resolution environment map to represent illumination, while Zhang *et al*. [76] uses MLPs to model indirect illumination. However, these methods require long training and inference times because of the multiple MLPs required for representation. In contrast, NVDIFFREC [39] uses deferred shading and the split-sum approximation for direct illumination but does not model self-shadows, which are crucial in hands. Further, NVDIFFRECMC [14] evaluates direct lighting integral using Monte

Carlo integration and ray tracing to model shadow rays. However, it requires multiview images of a static object. Our method follows the renderer from NVDIFFRECMC to account for self-shadows, however, unlike NVDIFFRECMC, our method is capable of reconstructing from a monocular video of a dynamic hand.

III. Hand Reconstruction and Tracking

Given a monocular RGB-D video of a user's hand performing arbitrary unknown poses under unknown real-world environment lighting, our goal is to reconstruct a personalized hand avatar representing the user's hand shape and appearance (with disentangled illumination) while tracking its articulated pose at each frame. Our hand avatar representation consists of a triangle mesh to capture the user's hand geometry and spatially varying materials (stored as 2D textures) to capture the user's hand appearance. We assume that the hand is illuminated under an unknown environment lighting, which we model using a high dynamic range (HDR) environment probe.

We propose a novel method, *HandRT*, to simultaneously reconstruct a personalized hand avatar while tracking its pose from a monocular RGB-D video. Our method jointly optimizes a parametric mesh, material, and lighting using a differentiable rendering framework integrated with articulated registration. Our method is summarized in Figure 2, and each step is described in detail below; Section III-A describes the parametric mesh used to model the hand geometry, Section III-B details the material and lighting model while Section III-B explains our deferred rendering pipeline. Finally, we explain our avatar reconstruction optimization in Section III-C.

A. Hand Shape Model

Our parametric mesh model is based on the MANO [49] hand model for geometry and HTML [47] for UV mapping. The MANO model is a parameterized human hand model described by a function of pose $\theta \in \mathbb{R}^{45}$ (capturing local rotation at each of the 15 joints) and shape $\beta \in \mathbb{R}^{10}$ (coefficients for the PCA shape blend shapes) returning $N = 778$ vertices and $F = 1538$ faces. Additionally, to account for the global rigid transformation, we include global orientation and translation in the pose resulting in $\theta \in \mathbb{R}^{45+6}$. However, the original MANO model is too coarse to capture detailed geometry and appearance. Inspired by Alldieck et al. [1], we adapt MANO as follows.

Initially, we allow offsets $\mathbf{D} \in \mathbb{R}^{3N}$ from the template $\mathbf{T} \in \mathbb{R}^{3N}$:

$$M(\beta, \theta, \mathbf{D}) = W(T(\beta, \theta, \mathbf{D}), J(\beta), \theta, \mathbf{W}) \quad (2)$$

$$T(\beta, \theta, \mathbf{D}) = \mathbf{T} + B_s(\beta) + B_p(\theta) + \mathbf{D} \quad (3)$$

where W is a linear blend-skinning function using the skinning weights $\mathbf{W}$ to pose the rest-pose skeleton joints $J(\beta)$ and vertices $T(\beta, \theta, \mathbf{D})$ (obtained after pose- $B_p(\theta)$ and shape-dependent $B_s(\beta)$ deformations).

Further, to capture fine-level details, we uniformly subdivide the template mesh by adding new vertices on edge midpoints and correspondingly update the parametric mesh model. Our subdivided mesh has $N = 14652$ vertices and $F = 38450$ faces after performing two iterations of subdivision. Our extended MANO is differentiable with respect to the geometry parameters shape β, pose θ, and offsets $\mathbf{D}$, thus enabling gradient backpropagation.

B. Hand Appearance Model

Material: We model the hand material's reflectance using Disney's principled BRDF [34], which is a parametric model offering a good balance between simplicity and flexibility. Specifically, our BRDF is parameterized by a three-channel diffuse albedo $\mathbf{k}_d$, a roughness channel r, a metalness channel m and a tangent space normal map $\mathbf{n}$. Since hand skin does not resemble a metal, we clamp the metalness value m to be low. The specular highlight color is calculated according to $\mathbf{k}_s = (1 - m) \cdot 0.04 + m \cdot \mathbf{k}_d$ [35]. Further, following a standard convention [39], we store these values in a texture $\mathbf{k}_{orm} = (o, r, m)$, where o is left unused. Our physically-based material model enables us to model the realistic hand appearance of various users.

Lighting: We model environment lighting using an HDR light probe, which supplies incident radiance from all directions on the sphere. Specifically, we use a 2D floating point texture map parameterized by spherical coordinates to represent our light probe. Our lighting representation enables us to model real-world environment lighting that is typically found in the usage scenarios for AR/VR devices.

Rendering: We now describe how the geometry, material, and lighting are utilized in a differentiable deferred rendering pipeline. We first rasterize the scene into geometry buffers (G-buffer) that include the surface intersection point and normal and the interpolated material parameters for each pixel. Given the G-buffer, we perform the shading pass to determine the outgoing radiance. Specifically, we use Monte Carlo (MC) integration to estimate the outgoing radiance. This requires us to calculate visibility, which is evaluated by tracing a shadow ray in the incoming light direction. This formulation allows us to explicitly handle self-shadows caused due to self-occlusion among fingers. More information about our Monte Carlo path tracing is provided in the supplementary material.

The rasterizer [28] and shading [14] are differentiable and thus allow gradients to be backpropagated to lighting, material as well as geometry parameters.

C. Optimization

Given a video with F frames, we jointly optimize the mesh parameters $\left(\beta, \mathbf{D}, \{\theta_f\}_{f=1}^{F}\right)$, material parameters $(\mathbf{k}_d, \mathbf{k}_s, \mathbf{n})$ and the light probe texture using self-supervision (analysis-by-synthesis) on the monocular RGB-D video. Our optimization minimizes the sum of depth and RGB data terms and various regularizer terms.

$$E_{\text{total}} = \omega_{\text{data}_{\text{depth}}} E_{\text{data}_{\text{depth}}} + \omega_{\text{data}_{\text{RGB}}} E_{\text{data}_{\text{RGB}}} + \sum_i \omega_i E_i \quad (4)$$

We describe each of these terms below.

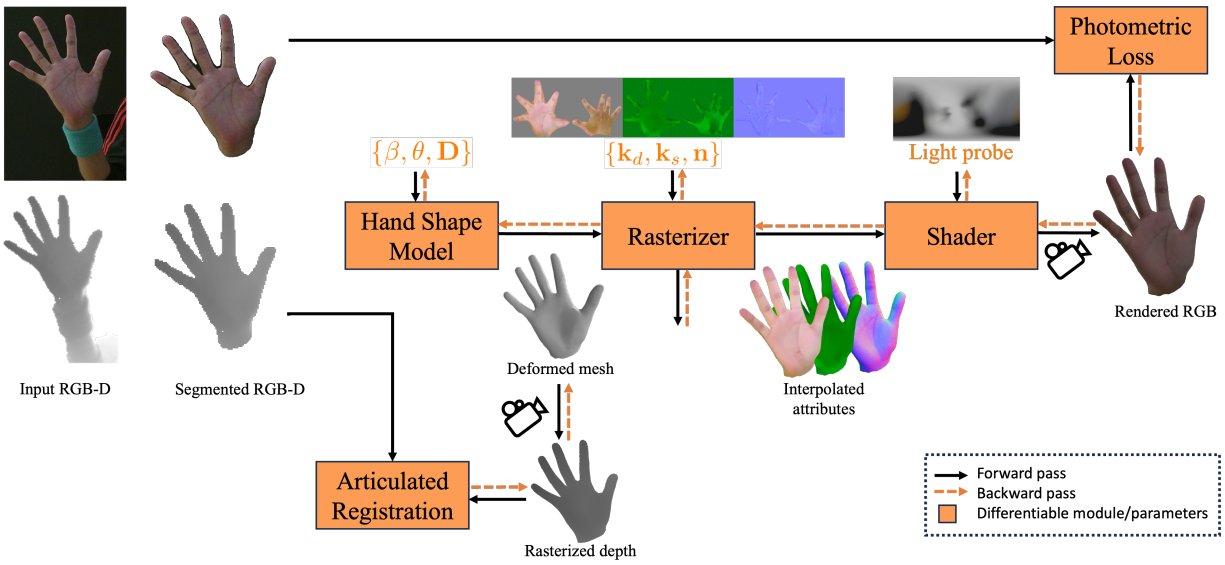

Fig. 2. We propose a novel method, *HandRT*, to reconstruct the personalized hand shape and appearance of a user's hand captured under real-world environment lighting while simultaneously tracking the articulated pose from a monocular RGB-D video. Our hand shape model is based on the MANO [49] hand model parameterized by shape β and pose θ which we further extend by subdividing and introducing offsets $\mathbf{D}$ to capture the detailed geometry. We use physically-based spatially-varying materials [34] $(\mathbf{k}_d, \mathbf{k}_s, \mathbf{n})$. We model lighting using a high dynamic range environment light probe to disentangle the intrinsic appearance of the user's hand and illumination. We use a differentiable deferred rendering pipeline: first, we rasterize the PBR material textures along with surface intersection point and normal to obtain interpolated vertex attributes and, then, use Monte Carlo integration to estimate outgoing radiance in the presence of an environment light. Our method jointly optimizes the parametric mesh, material, and lighting using differentiable rendering [21] integrated with our proposed articulated registration for tracking pose.

1) Depth data term: The depth data is used to minimize an articulated registration energy between the depth point cloud $x \in \mathbb{R}^{n_{3D} \times 3}$ and the mesh. (We use the term, articulated registration, since the rigid deformation is localized at each joint along the articulated kinematic chain.) We obtain the point cloud from the depth image f and apply *furthest point downsampling* [45] for efficient downstream processing. Additionally, we also calculate the normal $x_i^{\perp} \in \mathbb{R}^3$ to the i^{th} point by locally fitting a plane [51]. We use the point-to-plane ICP [9] to define the 3D data term

$$E_{\text{data}_{\text{depth}}}(\beta, \mathbf{D}, \theta_f) = \sum_{i=1}^{n_{3D}} (x_i - y_i(\beta, \mathbf{D}, \theta_f))^{\intercal} x_i^{\perp} \quad (5)$$

where $y_i \in \mathbb{R}^3$ is the closest point on the mesh corresponding to x_i, and $x_i^{\perp}$ is the surface normal at x_i.

Computing the closest point on the mesh for every data point is expensive since we have to project each point on every face and find the closest point on that triangle. Instead, we observe that our point cloud is obtained from a monocular depth sensor and thus use projection-based matching [50], i.e., we project points on the mesh according to the sensor view point, to obtain corresponding points by simply a look-up in the projected image space. Apart from being fast, this strategy prohibits selecting points that might be closer but face away from the camera. To increase robustness to outliers, we filter correspondences based on a threshold for Euclidean distance between positions and cosine similarity between normals. Finally, we choose source points along both directions (mesh to point cloud and point cloud to mesh) to further improve results when the overlap is small. These strategies improve the registration accuracy compared to a vanilla chamfer loss [3].

2) RGB data term: Our image space loss computes the L_1 norm of the error between the tone-mapped [39] (since our shading images are HDR) output rendered image $\tilde{\mathbf{c}}$ and the input reference image $\tilde{\mathbf{c}}_{ref}$, applied on tone-mapped versions. We also use a mask loss which computes a L_2 norm of the error between the silhouettes of the rendered $\mathbf{s}$ and input $\mathbf{s}_{ref}$ images. We obtain the input image silhouette $\mathbf{s}_{ref}$ using the mesh obtained during the initialization stage.

$$E_{\text{data}_{\text{RGB}}} = \|\tilde{\mathbf{c}} - \tilde{\mathbf{c}}_{ref}\|_1 + \|\mathbf{s} - \mathbf{s}_{ref}\|_2 \quad (6)$$

3) Geometry regularizer: We regularize the introduced offsets $\mathbf{D}$ by adding a Laplacian mesh regularizer [54]:

$$E_{\text{lap}} = \frac{1}{N} \sum_{i=1}^{N} \|\delta_i - \delta_i'\|^2 \quad (7)$$

where δ_i and δ_i' are the uniformly weighted differential of the optimized and the initial mesh for a vertex v_i given by $\delta_i = v_i - \frac{1}{|N_i|} \sum_{j \in N_i} v_j$, with N_i being the one-ring neighborhood of vertex v_i.

4) Material regularizer: We regularize material parameters using a smoothness loss similar to NeRFactor [75]. We define the smoothness prior for $\mathbf{k}_d$ as

$$E_{\mathbf{k}_d} = \sum_{y_{surf}} \|\mathbf{k}_d(y_{surf}) - \mathbf{k}_d(y_{surf} + \epsilon)\|^2 \quad (8)$$

where ϵ is a small random displacement vector, $\mathbf{k}_d(y_{surf})$ denotes the value of $\mathbf{k}_d$ at world space position y_{surf} at the

TABLE I. Weights used for each energy term.

$\omega_{\text{data}_{\text{depth}}}$	$\omega_{\text{data}_{\text{RGB}}}$	ω_{lap}	$\omega_{\mathbf{k}_d}$	$\omega_{\mathbf{k}_s}$	$\omega_{\mathbf{n}}$	ω_{LPIPS}	ω_{light}
0.5	1.0	1000	0.1	0.05	0.025	0.1	0.15

primary hit point on the object. We apply similar smoothness priors for specular $E_{\mathbf{k}_s}$ and normal $E_{\mathbf{n}}$ maps, respectively.

Additionally, we use the learned perceptual image patch similarity ($\mathcal{L}_{\text{LPIPS}}$) [74] loss to encourage perceptual similarity.

$$E_{\text{LPIPS}} = \mathcal{L}_{\text{LPIPS}}(\mathbf{c}, \mathbf{c}_{ref}) \tag{9}$$

5) Lighting regularizer: We note that the prior smoothness cannot disentangle material parameters and lighting. We use a monochrome image loss between demodulated lighting terms (i.e., before being multiplied by diffuse reflectance $\mathbf{k}_d$) and reference image $\mathbf{c}_{ref}$ [14].

$$E_{\text{light}} = \|Y(\mathbf{c}_d + \mathbf{c}_s) - V(\mathbf{c}_{ref})\|^2 \tag{10}$$

where $Y(\mathbf{x}) = (\mathbf{x}_r + \mathbf{x}_g + \mathbf{x}_b)/3$ is a luminance operator, and $V(\mathbf{x}) = \max(\mathbf{x}_r, \mathbf{x}_g, \mathbf{x}_b)$ is the HSV value component.

Implementation: We minimize E_{total} using SGD optimizer for frame-specific parameters $\left(\{\theta_f\}_{f=1}^{F}\right)$ and Adam optimizer [25] for other parameters. Our optimization strategy sequentially updates parameters for each frame, over multiple epochs, *i.e.* in each epoch (over all frames) we perform F update steps. We clamp the material parameters after each iteration to ensure physically valid textures. We use a higher learning rate for lighting parameters and a lower learning rate for material parameters. The values of the weights used in the regularization terms of our energy term are mentioned in Table I. We use PyTorch [43] for optimization. Given a monocular video with 100 frames of resolution 1024×1024, our method with 100 epochs can reconstruct a reasonable hand avatar within 10 minutes on an NVIDIA GeForce RTX 3080 Ti.

Initialization: We initialize the pose in the first frame by minimizing the L_1 distance of the detected 2D hand joint positions $\{j_i'\}_i^{21}$ (obtained using MediaPipe [63]) and the projected 2D joint positions $\{j_i\}_i^{21}$ of the posed mesh.

$$E_{\text{2D}}(\theta) = \sum_{i=1}^{21} \|j_i(\theta) - j_i'\|^2 \tag{11}$$

To initialize the pose for the subsequent frames, we use the previous frame's optimized pose. Notice the difference with Intrinsic Hand Avatar which requires detected 2D joint positions at each frame to estimate a coarse hand pose, whereas, we only require it for the first frame. Additionally, we initialize the shape using the mean PCA shape and randomly initialize the material and lighting parameters.

Our data term assumes the hand is segmented in the input image. To this end, we use the Segment Anything Model [26] to segment the hand region from the body and other background. Specifically, we use the projected 2D joint positions from the initialized mesh to obtain point prompts for segmentation. Next, we remove the forearm region using mesh annotations in the dataset or a wristband in our captured dataset.

IV. Experimental Results

We evaluate our method on the task of hand avatar reconstruction and tracking and compare it with state-of-the-art methods. Finally, to highlight the impact of our method, we demonstrate applications in novel pose synthesis and relighting in the supplementary.

A. Datasets

We compare our method with state-of-the-art hand avatar reconstruction and hand mesh recovery methods on the following public datasets, along with our captured dataset described below.

a) InterHand2.6M: To compare with RGB-based avatar reconstruction methods, we use the InterHand2.6M dataset [37]. This data is captured in a dome with controlled lighting and provides fitted MANO parameters obtained using NeuralAnnot [38]. The images are centered, scaled by a factor of 1.3 around the hand, and cropped to 256×256.

b) GuessWho: To compare with depth-based hand shape recovery methods, we use the GuessWho [61] dataset. The data is acquired from twelve different users performing in front of an Intel RealSense SR300 camera (a consumer-level time-of-flight depth sensor).

c) Captured: Further, to evaluate the method on various hand shapes and appearances, we capture our dataset using a calibrated (known camera intrinsics) RGB-D camera, Kinect v2 [29] recorded at 30 frames per second, with RGB resolution of 1920×1080 and depth resolution of 512×424. We ask the user to start with a stretched hand pose and perform arbitrary poses showing all regions of their hand (see Figure 1). Our dataset comprises approximately 10000 frames captured from 7 different users. This dataset is challenging because of the complex poses and real-world lighting resembling the actual scenario for AR/VR applications.

B. Metrics

We report metrics focusing on the rendered image quality, including PSNR, SSIM [66], and LPIPS, along with intersection-over-union (IoU) to evaluate geometry silhouette. Due to the absence of ground-truth meshes for real data, we cannot explicitly evaluate geometry reconstruction. However, the above metrics implicitly capture geometry reconstruction together with appearance reconstruction.

C. Evaluation of Hand Tracking

1) RGB-based Methods: We evaluate the accuracy of our mesh recovery on our captured dataset by comparing it with state-of-the-art hand shape and pose estimation methods: ACR [71] that regresses the MANO parameters for pose and MeshGraphormer [32] which uses a GraphCNN to regress mesh vertices directly. Both methods were trained on the FreiHAND dataset [80], while ACR was additionally trained on a manually obtained in-the-wild dataset. In Fig. 3, we observe that our method consistently produces a tighter fit on different poses by a large margin. We also report the intersection-over-union (IoU) metric in Table II and observe that the accuracy of our method is significantly better.

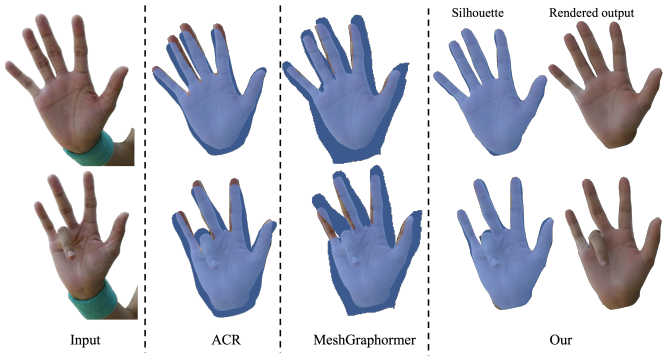

Fig. 3. Qualitative comparison of our method with state-of-the-art hand mesh reconstruction methods on our captured dataset. We composite the recovered mesh (blue) over the segmented input image. Notice the accurate overlap of our recovered mesh with the segmented input image.

TABLE II. Quantitative evaluation of hand mesh recovery on our captured data.

Method	Input Data	IoU↑
ACR [71]	RGB	0.742
MeshGraphormer [32]	RGB	0.622
Ours	RGB-D	**0.958**

2) Depth-based Methods: We evaluate the accuracy of our RGB-D hand tracking with generative depth-based hand tracking methods HAdjust [48] and HOnline [61] on the GuessWho dataset in Figure 4. We observe that the RGB input in our method provides helpful shading cues for accurate registration.

D. Evaluation of Hand Appearance Reconstruction

We compare our method with previous hand avatar reconstruction methods HARP [22] and HandAvatar [7] on the InterHand2.6M dataset. As this dataset contains only RGB input, we skip the depth data term while retaining other terms in our optimization, thus highlighting our method's versatility to handle different input modality. Unlike our physically realistic shading, HARP uses shadow mapping in its differentiable rendering pipeline and only supports a single light source. In Figure 5, we observe that our method produces sharper texture details and handles shadows more accurately than HARP. We also observe that our method accurately captures fine geometric details as seen in the shading images in Figure 7 whereas these details are missing in the shading images of HandAvatar. We also see that our method produces correct shadows when compared to the self-occlusion-aware shading field in HandAvatar.

We further evaluate our rendering result with recent monocular methods using volume rendering (HumanNeRF [67]) and surface rendering (SelfRecon [16]) in Figure 6. Unlike our fast test-time optimization (minutes), these methods require long training times (hours).

We quantitatively evaluate our method and compare it with prior methods in Table III and Table IV on the InterHand2.5M dataset. We observe that our method outperforms existing methods on all metrics.

TABLE III. Quantitative evaluation on InterHand2.6M 5fps dataset.

Method	Input Data	PSNR↑	SSIM↑	LPIPS↓
SelfRecon [16]	RGB	26.38	0.878	0.142
HumanNeRF [67]	RGB	27.64	0.883	0.114
Ours	RGB	**28.66**	**0.897**	**0.090**

TABLE IV. Quantitative evaluation on InterHand2.6M 30fps dataset.

Method	Input Data	PSNR↑	SSIM↑	LPIPS↓
HARP [22]	RGB	16.157	0.866	0.167
HandAvatar [7]	RGB	29.423	0.914	0.088
Ours	RGB	**31.179**	**0.936**	**0.061**

Finally, we show the output of our method on our captured dataset in Figure 8. We observe that our method accurately reconstructs detailed textures and disentangles lighting from albedo.

E. Ablation

1) HandRT v/s Intrinsic Hand Avatar: We evaluate the accuracy of reconstruction using the RGB-based Intrinsic Hand Avatar and our RGB-D-based HandRT on our captured dataset in Figure 9. Additionally, we quantitatively compare HandRT with Intrinsic Hand Avatar in Table V. We also used the optimized pose from HandRT as initialization in Intrinsic Hand Avatar to demonstrate the accuracy of pose estimation in HandRT and also to highlight the importance of accurate pose in avatar reconstruction (row 2 in Table V).

2) Effect of Lighting in Tracking: The shading information obtained by modeling environment lighting is essential to recover high-fidelity shape, including flexion creases, as shown in Fig. 10.

V. Discussion

Impact: Our method takes an important step in reconstructing an intrinsic hand avatar of a user's hand from a monocular RGB-D video. Our method addresses one of the main issues of other avatar reconstruction methods which require initial coarse pose, do not disentangle material and lighting, and fail to generalize to new users. Thus, AR/VR applications can leverage our method to reconstruct a user-specific hand avatar capturing the user's hand shape and appearance from unconstrained poses and provide a more natural and immersive experience.

Limitations: If the estimated pose is incorrect, it leads to incorrect appearance estimation. For example, in a fist pose, the thumb is incorrectly estimated to be under the index/middle finger, whereas it is above them in the input groundtruth. As a result, our method confuses the recovered texture of the

TABLE V. Quantitative comparison with our previous method Intrinsic Hand Avatar (IHA) [21] on our captured dataset.

Method	Input Data	PSNR↑	SSIM↑	LPIPS↓
IHA [21]	RGB	27.83	0.816	0.117
IHA [21] (pose from HandRT)	RGB	29.53	0.920	0.069
Ours	RGB-D	**29.81**	**0.934**	**0.067**

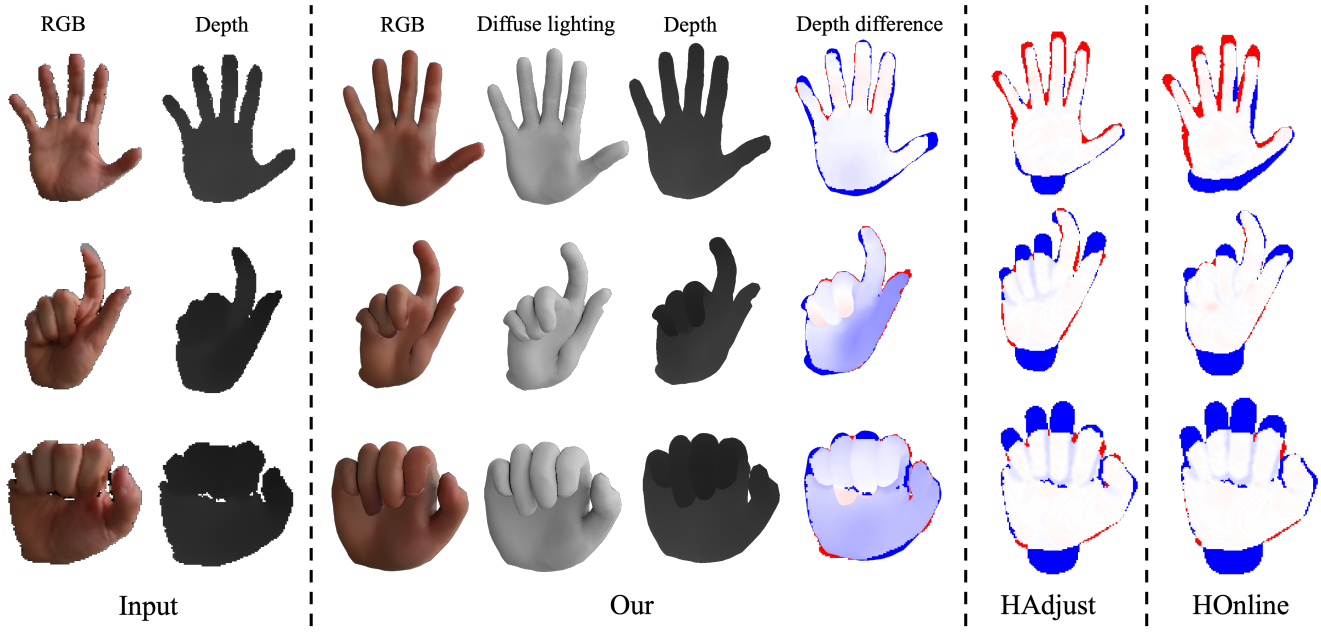

Fig. 4. Qualitative comparison of our self-supervised RGB-D tracking with generative depth-based hand tracking methods HAdjust [48] and HOnline [61] on the GuessWho dataset.

Fig. 5. We compare the reconstructed hand obtained from our method with HARP [22], which uses a shadow-aware differentiable rendering in an analysis-by-synthesis approach. Unlike the blurred output from HARP, our method accurately captures fine-level appearance details.

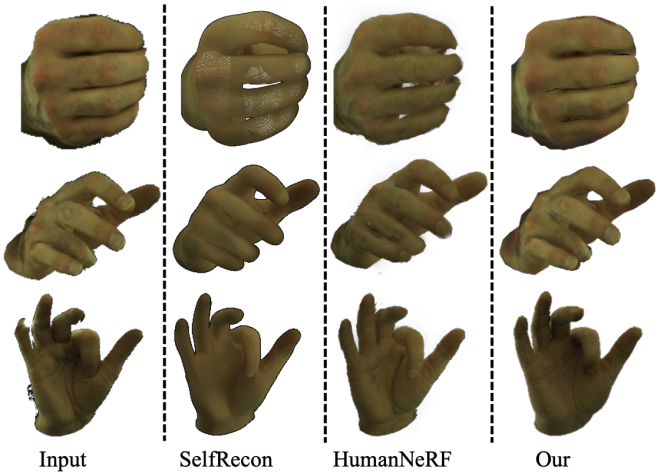

Fig. 6. Our method reconstructs high-fidelity texture compared to the surface rendering method of SelfRecon [16] and correctly captures illumination compared to the volume rendering method of HumanNeRF [67].

index/middle fingers with that of the thumb, eventually leading to incorrect rendering as shown in Figure 11.

Further, our method assumes that the hand in the input video is clearly visible and thus cannot handle occlusion caused by other objects in the scene. If any part of the user's hand is never visible in the video, that region in the reconstructed hand avatar will remain at its initialized value as it never gets updated in the optimization. Although our method can handle dynamic hand poses in the video, it assumes static lighting throughout the video. We clamp the metalness of our material to lower values, which does not allow the modeling of shiny fingernails.

Future work: The material model can be improved by incorporating subsurface scattering observed in human skin. Incorporating bulges and wrinkles into the geometry deformation model can greatly improve realism and enable capturing fine-level shape and appearance details. Another interesting direction would be to model fine hair on the back of the hand to improve the realism of the avatar.

Ethical Considerations: The recent advances in reconstructing digital avatars raise the concern of nefarious use cases. The foremost danger in reconstructing such a hand avatar from an RGB camera is the use of fingerprints. To protect users' fingerprint privacy, all images in the paper are down-sampled before the acquisition, which also leads to slightly blurry textures.

VI. Conclusion

We present a method for reconstructing the hand shape and appearance of a user's hand captured under real-world environment lighting while simultaneously tracking the articulated pose from monocular RGB-D video. In contrast to existing methods which require known initial coarse pose, our method can leverage depth data and utilize tracking to obtain

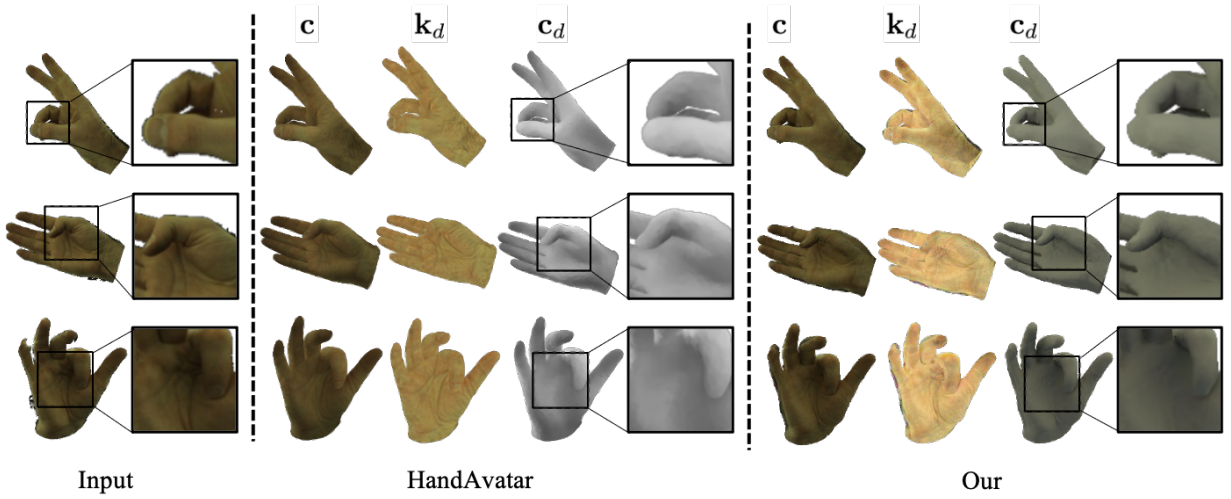

Fig. 7. We compare the shadow-aware capability of our method with HandAvatar, which uses a self-occlusion-aware shading field to model self-shadows. Compared to the local-pair occupancy field of HandAvatar, our method explicitly models shadow rays to produce more accurate self-shadows in complicated poses.

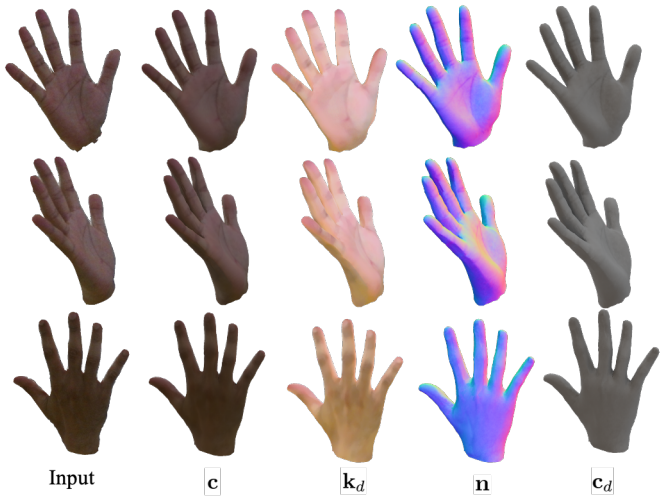

Fig. 8. Reconstruction from images of our captured dataset (user1), we can see fine geometric details captured on the palm in the normal and shading images.

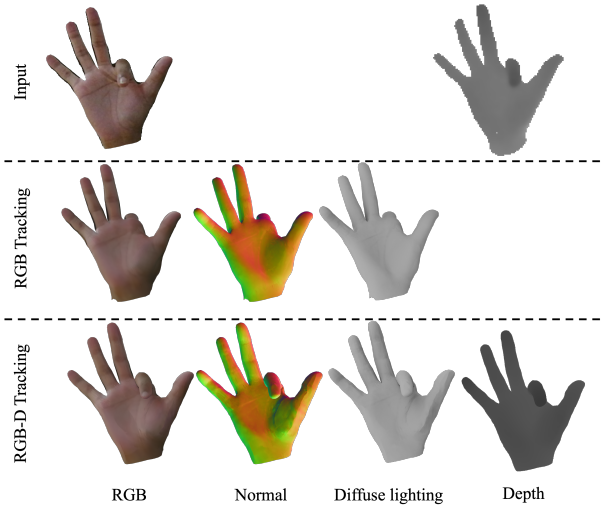

Fig. 9. Qualitative comparison of our self-supervised tracking method with RGB and RGB-D input on our captured dataset. Without depth input, we observe that the index finger incorrectly bends backward, which minimizes the projected 2D landmark loss.

highly accurate poses. Also, unlike previous methods where illumination is baked into the appearance, our method disentangles the intrinsic properties of the underlying appearance and environment lighting, leading to realistic self-shadows. Further, our method is capable of precisely tracking the reconstructed personalized avatar from either RGB or RGB-D input modality. Our self-supervised approach optimizes using only the input image observations, thus generalizing to unseen poses. Our method can generate an avatar within minutes and supports arbitrary hand poses captured in a real-world environment, which enables our method to be used for a variety of applications.

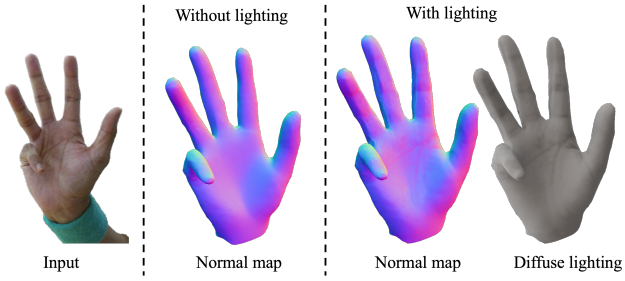

Fig. 10. Our lighting model enables capturing fine-level geometric details like flexion creases and handles self-shadows.

References

[1] Thiemo Alldieck, Marcus Magnor, Weipeng Xu, Christian Theobalt, and Gerard Pons-Moll. Detailed human

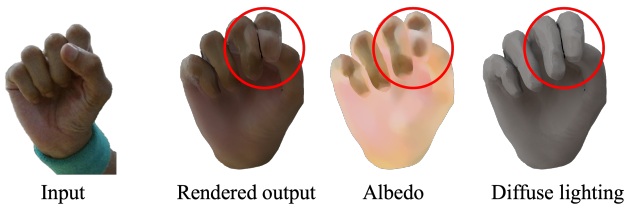

Fig. 11. Due to incorrect estimated pose, our method confuses the recovered texture of the index/middle fingers with that of the thumb, eventually leading to incorrect rendering.

avatars from monocular video. In *International Conference on 3D Vision*, 2018. 4

[2] Seungryul Baek, Kwang In Kim, and Tae-Kyun Kim. Pushing the envelope for rgb-based dense 3d hand pose estimation via neural rendering. In *CVPR*, 2019. 2

[3] Harry G Barrow, Jay M Tenenbaum, Robert C Bolles, and Helen C Wolf. Parametric correspondence and chamfer matching: Two new techniques for image matching. In *IJCAI*, 1977. 5

[4] Volker Blanz and Thomas Vetter. A morphable model for the synthesis of 3d faces. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques*, 1999. 3

[5] Mark Boss, Raphael Braun, Varun Jampani, Jonathan T. Barron, Ce Liu, and Hendrik P.A. Lensch. Nerd: Neural reflectance decomposition from image collections. In *ICCV*, 2021. 3

[6] Adnane Boukhayma, Rodrigo de Bem, and Philip HS Torr. 3d hand shape and pose from images in the wild. In *CVPR*, 2019. 1, 2

[7] Xingyu Chen, Baoyuan Wang, and Heung-Yeung Shum. Hand avatar: Free-pose hand animation and rendering from monocular video. In *CVPR*, 2023. 1, 2, 3, 7

[8] Xu Chen, Yufeng Zheng, Michael J Black, Otmar Hilliges, and Andreas Geiger. Snarf: Differentiable forward skinning for animating non-rigid neural implicit shapes. In *ICCV*, 2021. 3

[9] Yang Chen and Gerard Medioni. Object modeling by registration of multiple range images. In *ICRA*, 1991. 5

[10] Enric Corona, Tomas Hodan, Minh Vo, Francesc Moreno-Noguer, Chris Sweeney, Richard Newcombe, and Lingni Ma. Lisa: Learning implicit shape and appearance of hands. In *CVPR*, 2022. 1, 3

[11] Martin de La Gorce, David J. Fleet, and Nikos Paragios. Model-based 3d hand pose estimation from monocular video. *IEEE TPAMI*, 33(9), 2011. 2

[12] Yao Feng, Jinlong Yang, Marc Pollefeys, Michael J. Black, and Timo Bolkart. Capturing and animation of body and clothing from monocular video. In *SIGGRAPH Asia 2022 Conference Papers*, 2022. 1, 3

[13] Xuan Gao, Chenglai Zhong, Jun Xiang, Yang Hong, Yudong Guo, and Juyong Zhang. Reconstructing personalized semantic facial nerf models from monocular video. *ACM TOG*, 41(6), 2022. doi: 10.1145/3550454.3555501. 1

[14] Jon Hasselgren, Nikolai Hofmann, and Jacob Munkberg. Shape, light, and material decomposition from images using monte carlo rendering and denoising. In *NeurIPS*, 2022. 3, 4, 6

[15] Weiting Huang, Pengfei Ren, Jingyu Wang, Qi Qi, and Haifeng Sun. Awr: Adaptive weighting regression for 3d hand pose estimation. In *AAAI*, 2020. 3

[16] Boyi Jiang, Yang Hong, Hujun Bao, and Juyong Zhang. Selfrecon: Self reconstruction your digital avatar from monocular video. In *CVPR*, 2022. 1, 3, 7, 8

[17] Tianjian Jiang, Xu Chen, Jie Song, and Otmar Hilliges. Instantavatar: Learning avatars from monocular video in 60 seconds. In *CVPR*, 2023. 1, 3

[18] James T. Kajiya. The rendering equation. In *Proceedings of the 13th Annual Conference on Computer Graphics and Interactive Techniques*, pages 143–150, 1986. 3

[19] Pratik Kalshetti and Parag Chaudhuri. Local Scale Adaptation to Hand Shape Model for Accurate and Robust Hand Tracking. *Computer Graphics Forum (ACM SIGGRAPH/Eurographics Symposium on Computer Animation)*, 41:219–229, 2022. 1

[20] Pratik Kalshetti and Parag Chaudhuri. Local scale adaptation for augmenting hand shape models. In *ACM SIGGRAPH 2022 Posters*, 2022. 1

[21] Pratik Kalshetti and Parag Chaudhuri. Intrinsic hand avatar: Illumination-aware hand appearance and shape reconstruction from monocular rgb video. In *WACV*, 2024. 2, 5, 7

[22] Korrawe Karunratanakul, Sergey Prokudin, Otmar Hilliges, and Siyu Tang. Harp: Personalized hand reconstruction from a monocular rgb video. In *CVPR*, 2023. 1, 2, 3, 7, 8

[23] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM TOG*, 42(4), 2023. 3

[24] Sameh Khamis, Jonathan Taylor, Jamie Shotton, Cem Keskin, Shahram Izadi, and Andrew Fitzgibbon. Learning an efficient model of hand shape variation from depth images. In *CVPR*, 2015. 3

[25] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015. 6

[26] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, and Ross Girshick. Segment anything. In *ICCV*, 2023. 6

[27] Dominik Kulon, Riza Alp Guler, Iasonas Kokkinos, Michael M. Bronstein, and Stefanos Zafeiriou. Weakly-supervised mesh-convolutional hand reconstruction in the wild. In *CVPR*, 2020. 1, 2

[28] Samuli Laine, Janne Hellsten, Tero Karras, Yeongho Seol, Jaakko Lehtinen, and Timo Aila. Modular primitives for high-performance differentiable rendering. *ACM TOG*, 39(6), 2020. 4

[29] Microsoft Learn. Kinect for windows. https://learn.microsoft.com/en-us/windows/apps/design/devices/kinect-for-windows, 2022. Accessed: 2023-11-30. 6

[30] Yuwei Li, Longwen Zhang, Zesong Qiu, Yingwenqi

Jiang, Nianyi Li, Yuexin Ma, Yuyao Zhang, Lan Xu, and Jingyi Yu. Nimble: A non-rigid hand model with bones and muscles. *ACM TOG*, 41(4), 2022. 1, 3

[31] Kevin Lin, Lijuan Wang, and Zicheng Liu. End-to-end human pose and mesh reconstruction with transformers. In *CVPR*, 2021. 1, 2

[32] Kevin Lin, Lijuan Wang, and Zicheng Liu. Mesh graphormer. In *ICCV*, 2021. 1, 2, 6, 7

[33] Alexandros Makris and Antonis Argyros. Model-based 3d hand tracking with on-line shape adaptation. In *BMVC*, 2015. 2

[34] Stephen McAuley, Stephen Hill, Naty Hoffman, Yoshiharu Gotanda, Brian Smits, Brent Burley, and Adam Martinez. Practical physically-based shading in film and game production. In *ACM SIGGRAPH 2012 Courses*, 2012. 1, 4, 5

[35] Stephen McAuley, Stephen Hill, Adam Martinez, Ryusuke Villemin, Matt Pettineo, Dimitar Lazarov, David Neubelt, Brian Karis, Christophe Hery, Naty Hoffman, and Hakan Zap Andersson. Physically based shading in theory and practice. In *ACM SIGGRAPH 2013 Courses*, 2013. 4

[36] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020. 3

[37] Gyeongsik Moon, Shoou-I Yu, He Wen, Takaaki Shiratori, and Kyoung Mu Lee. Interhand2.6m: A dataset and baseline for 3d interacting hand pose estimation from a single rgb image. In *ECCV*, 2020. 6

[38] Gyeongsik Moon, Hongsuk Choi, and Kyoung Mu Lee. Neuralannot: Neural annotator for 3d human mesh training sets. In *CVPRW*, 2022. 6

[39] Jacob Munkberg, Jon Hasselgren, Tianchang Shen, Jun Gao, Wenzheng Chen, Alex Evans, Thomas Müller, and Sanja Fidler. Extracting Triangular 3D Models, Materials, and Lighting From Images. In *CVPR*, 2022. 3, 4, 5

[40] Michael Oechsle, Songyou Peng, and Andreas Geiger. Unisurf: Unifying neural implicit surfaces and radiance fields for multi-view reconstruction. In *ICCV*, 2021. 3

[41] Iason Oikonomidis, Nikolaos Kyriazis, and Antonis A Argyros. Efficient model-based 3d tracking of hand articulations using kinect. In *BMVC*, 2011. 2

[42] Keunhong Park, Utkarsh Sinha, Jonathan T. Barron, Sofien Bouaziz, Dan B Goldman, Steven M. Seitz, and Ricardo Martin-Brualla. Nerfies: Deformable neural radiance fields. *ICCV*, 2021. 3

[43] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. 2019. 6

[44] Chandradeep Pokhariya, Ishaan N Shah, Angela Xing, Zekun Li, Kefan Chen, Avinash Sharma, and Srinath Sridhar. Manus: Markerless hand-object grasp capture using articulated 3d gaussians. In *CVPR*, 2024. 3

[45] Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *CVPR*, 2017. 5

[46] Chen Qian, Xiao Sun, Yichen Wei, Xiaoou Tang, and Jian Sun. Realtime and robust hand tracking from depth. In *CVPR*, 2014. 3

[47] Neng Qian, Jiayi Wang, Franziska Mueller, Florian Bernard, Vladislav Golyanik, and Christian Theobalt. Html: A parametric hand texture model for 3d hand reconstruction and personalization. In *ECCV*, 2020. 1, 3, 4

[48] Edoardo Remelli, Anastasia Tkach, Andrea Tagliasacchi, and Mark Pauly. Low-dimensionality calibration through local anisotropic scaling for robust hand model personalization. In *ICCV*, 2017. 7, 8

[49] Javier Romero, Dimitrios Tzionas, and Michael J. Black. Embodied hands: Modeling and capturing hands and bodies together. *ACM TOG*, 36(6):245:1–245:17, 2017. 1, 2, 4, 5

[50] Szymon Rusinkiewicz and Marc Levoy. Efficient variants of the icp algorithm. In *Proceedings Third International Conference on 3-D Digital Imaging and Modeling*, 2001. 5

[51] Radu Bogdan Rusu. *Semantic 3D Object Maps for Everyday Manipulation in Human Living Environments*. PhD thesis, Computer Science department, Technische Universitaet Muenchen, Germany, October 2009. 5

[52] Toby Sharp, Cem Keskin, Duncan Robertson, Jonathan Taylor, Jamie Shotton, David Kim, Christoph Rhemann, Ido Leichter, Alon Vinnikov, Yichen Wei, Daniel Freedman, Eyal Krupka, Andrew Fitzgibbon, Shahram Izadi, and Pushmeet Kohli. Accurate, robust, and flexible real-time hand tracking. In *CHI*, 2015. 3

[53] Jingjing Shen, Thomas J. Cashman, Qi Ye, Tim Hutton, Toby Sharp, Federica Bogo, Andrew Fitzgibbon, and Jamie Shotton. The phong surface: Efficient 3d model fitting using lifted optimization. In *ECCV*, 2020. 1, 3

[54] O. Sorkine, D. Cohen-Or, Y. Lipman, M. Alexa, C. Rössl, and H.-P. Seidel. Laplacian surface editing. In *Eurographics/ACM SIGGRAPH Symposium on Geometry Processing*, 2004. 5

[55] Shih-Yang Su, Frank Yu, Michael Zollhöfer, and Helge Rhodin. A-nerf: Articulated neural radiance fields for learning human shape, appearance, and pose. In *NeurIPS*, 2021. 1, 3

[56] Xiao Sun, Yichen Wei, Shuang Liang, Xiaoou Tang, and Jian Sun. Cascaded hand pose regression. In *CVPR*, 2015. 3

[57] Andrea Tagliasacchi, Matthias Schröder, Anastasia Tkach, Sofien Bouaziz, Mario Botsch, and Mark Pauly. Robust articulated-icp for real-time hand tracking. *Computer Graphics Forum*, 34:5:101–114, 2015. 2

[58] David Joseph Tan, Tom Cashman, Jonathan Taylor, Andrew Fitzgibbon, Daniel Tarlow, Sameh Khamis, Shahram Izadi, and Jamie Shotton. Fits like a glove: Rapid and reliable hand shape personalization. In *CVPR*,

2016. 1, 3

[59] Jonathan Taylor, Richard Stebbing, Varun Ramakrishna, Cem Keskin, Jamie Shotton, Shahram Izadi, , and Andrew Fitzgibbon. User-specific hand modeling from monocular depth sequences. In *CVPR*, 2014. 1, 3

[60] Jonathan Taylor, Lucas Bordeaux, Thomas Cashman, Bob Corish, Cem Keskin, Eduardo Soto, David Sweeney, Julien Valentin, Benjamin Luff, Arran Topalian, Erroll Wood, Sameh Khamis, Pushmeet Kohli, Toby Sharp, Shahram Izadi, Richard Banks, Andrew Fitzgibbon, and Jamie Shotton. Efficient and precise interactive hand tracking through joint, continuous optimization of pose and correspondences. *ACM TOG*, 35, 2016. 3

[61] Anastasia Tkach, Andrea Tagliasacchi, Edoardo Remelli, Mark Pauly, and Andrew Fitzgibbon. Online generative model personalization for hand tracking. *ACM TOG*, 36 (6):1–11, 2017. 6, 7, 8

[62] Jonathan Tompson, Murphy Stein, Yann Lecun, and Ken Perlin. Real-time continuous pose recovery of human hands using convolutional networks. *ACM TOG*, 33, 2014. 3

[63] Andrey Vakunov, Chuo-Ling Chang, Fan Zhang, George Sung, Matthias Grundmann, and Valentin Bazarevsky. Mediapipe hands: On-device real-time hand tracking. In *Workshop on Computer Vision for AR/VR*, 2020. 6

[64] C. Wan, T. Probst, L. Gool, and A. Yao. Dense 3d regression for hand pose estimation. In *CVPR*, 2018. 3

[65] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. Neus: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. *NeurIPS*, 2021. 3

[66] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE TIP*, 13(4):600–612, 2004. 6

[67] Chung-Yi Weng, Brian Curless, Pratul P. Srinivasan, Jonathan T. Barron, and Ira Kemelmacher-Shlizerman. HumanNeRF: Free-viewpoint rendering of moving people from monocular video. In *CVPR*, 2022. 1, 7, 8

[68] Fu Xiong, Boshen Zhang, Yang Xiao, Zhiguo Cao, Taidong Yu, Joey Zhou Tianyi, and Junsong Yuan. A2j: Anchor-to-joint regression network for 3d articulated pose estimation from a single depth image. In *ICCV*, 2019. 3

[69] Lior Yariv, Jiatao Gu, Yoni Kasten, and Yaron Lipman. Volume rendering of neural implicit surfaces. In *NeurIPS*, 2021. 3

[70] Alex Yu, Ruilong Li, Matthew Tancik, Hao Li, Ren Ng, and Angjoo Kanazawa. PlenOctrees for real-time rendering of neural radiance fields. In *ICCV*, 2021. 3

[71] Zhengdi Yu, Shaoli Huang, Chen Fang, Toby P. Breckon, and Jue Wang. Acr: Attention collaboration-based regressor for arbitrary two-hand reconstruction. In *CVPR*, 2023. 2, 6, 7

[72] Shanxin Yuan, Qi Ye, Björn Stenger, Siddhant Jain, and Tae-Kyun Kim. Bighand2.2m benchmark: Hand pose dataset and state of the art analysis. In *CVPR*, 2017. 3

[73] Kai Zhang, Fujun Luan, Qianqian Wang, Kavita Bala, and Noah Snavely. PhySG: Inverse rendering with spherical gaussians for physics-based material editing and relighting. In *CVPR*, 2021. 3

[74] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 6

[75] Xiuming Zhang, Pratul P. Srinivasan, Boyang Deng, Paul Debevec, William T. Freeman, and Jonathan T. Barron. Nerfactor: Neural factorization of shape and reflectance under an unknown illumination. *ACM TOG*, 40(6), 2021. 3, 5

[76] Yuanqing Zhang, Jiaming Sun, Xingyi He, Huan Fu, Rongfei Jia, and Xiaowei Zhou. Modeling indirect illumination for inverse rendering. In *CVPR*, 2022. 3

[77] Zhaohui Zhang, Shipeng Xie, Mingxiu Chen, and Haichao Zhu. Handaugment: A simple data augmentation method for depth-based 3d hand pose estimation. *arXiv*, 2020. 3

[78] Hao Zhao, Jinsong Zhang, Yu-Kun Lai, Zerong Zheng, Yingdi Xie, Yebin Liu, and Kun Li. High-fidelity human avatars from a single rgb camera. In *CVPR*, 2022. 3

[79] Yufeng Zheng, Victoria Fernández Abrevaya, Marcel C. Bühler, Xu Chen, Michael J. Black, and Otmar Hilliges. I M Avatar: Implicit morphable head avatars from videos. In *CVPR*, 2022. 1, 3

[80] Christian Zimmermann, Duygu Ceylan, Jimei Yang, Bryan Russel, Max Argus, and Thomas Brox. Freihand: A dataset for markerless capture of hand pose and shape from single rgb images. In *ICCV*, 2019. 2, 6

Pratik Kalshetti Pratik Kalshetti is an M.Tech.+Ph.D. dual degree student in the Department of Computer Science and Engineering at IIT Bombay, advised by Prof. Parag Chaudhuri. Previously, he completed his B.Tech. in Computer Science from IIT Jodhpur. Pratik has achieved a top-three place twice in the ACM Student Research Competition at SIGGRAPH (2019, 2022). His research on hand shape modeling was featured on the ACM SIGGRAPH blog. He served as a SIGGRAPH student volunteer in 2018 and 2021.

Parag Chaudhuri Parag Chaudhuri is a professor in the Department of Computer Science and Engineering at IIT Bombay. His current research centers around the simulation of virtual characters and natural phenomena to populate virtual worlds. His team develops methods to create, interact with and understand visual data. He was a postdoctoral researcher at MIRALab, University of Geneva and he holds a Ph.D. from IIT Delhi. He loves teaching Computer Graphics to enthusiastic students.